

9.3.1

a)

The system is periodic, but it is not chaotic because it's not sensitive to initial conditions, which is one of the requirements for chaos.

b)

In fact, the error never changes.

$$\begin{aligned}\dot{\theta}_1 &= w_1 \Rightarrow \theta_1(t) = w_1 t + C_1, & \theta_1(0) &= C_1 \\ \dot{\theta}_2 &= w_2 \Rightarrow \theta_2(t) = w_2 t + C_2, & \theta_2(0) &= C_2\end{aligned}$$

Adding error to the initial conditions makes a blob around the initial real initial condition, but the blob moves together and doesn't distort.

The system can be interpreted as movement of a dot on a sphere. w_1 controls how fast the dot moves in the latitudinal direction and w_2 how fast the dot moves in the longitudinal direction.

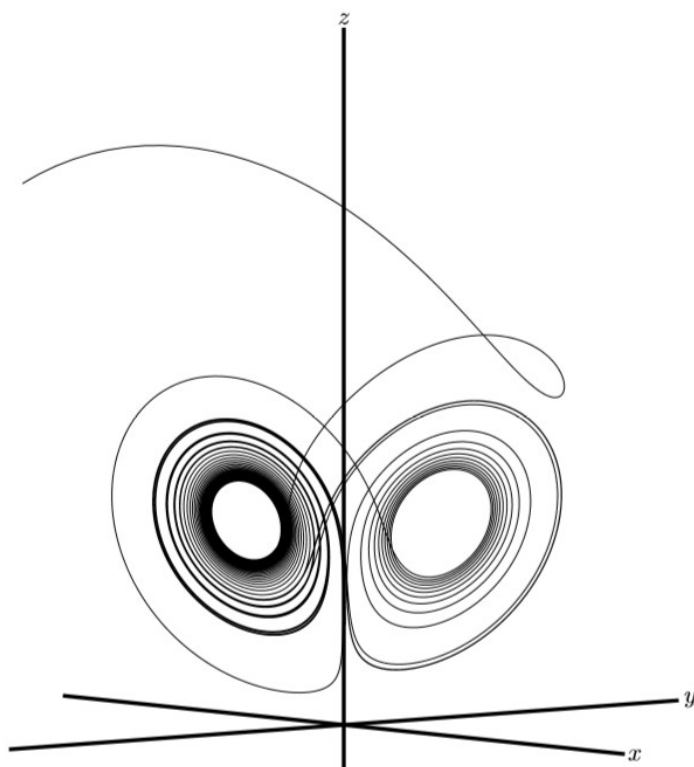
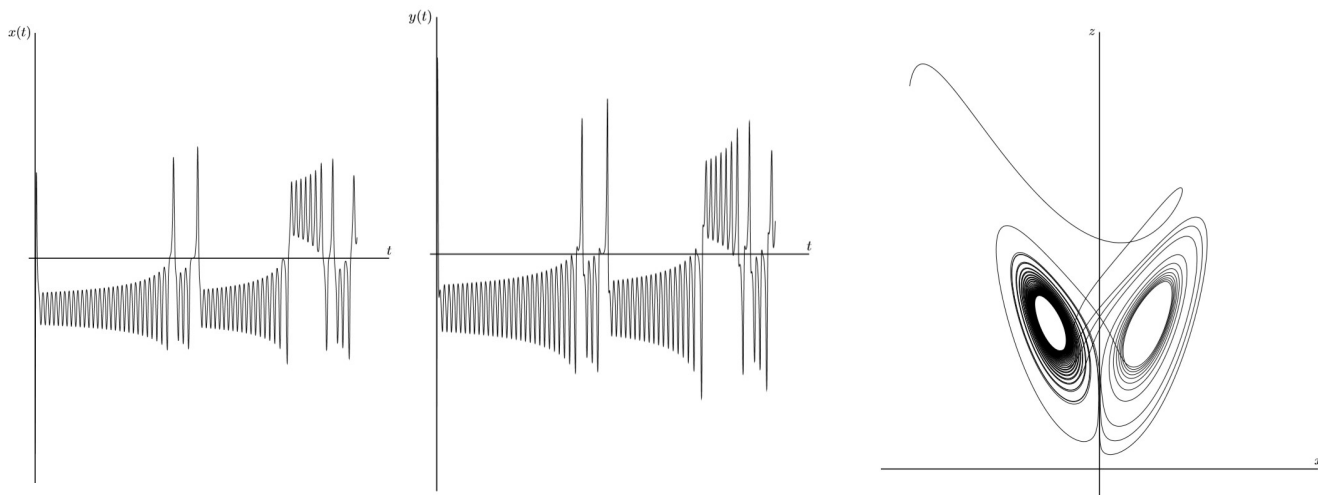
$$||(\vec{\theta}(t) + \vec{\delta}(t)) - \vec{\theta}(t)|| = ||\vec{\delta}(t)|| = ||(\delta_1, \delta_2)|| = ||(\delta_1, \delta_2)||e^{0t}$$

So the largest Liapunov exponent is 0.

9.3.4

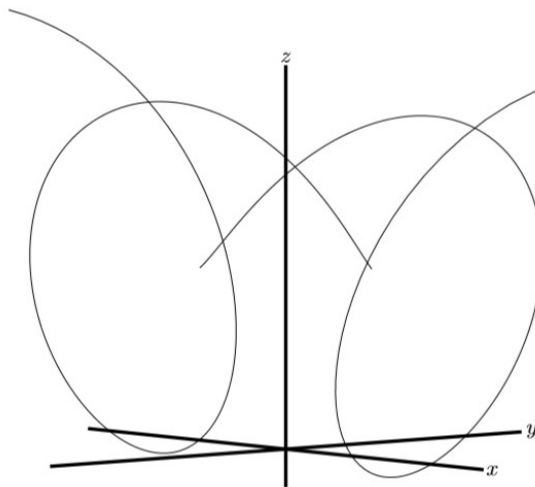
With the parameters $r = 24.5$ $\sigma = 10$ $b = \frac{8}{3}$ the trajectories either stay chaotic or go to one of two stable fixed points.

This simulation has initial condition $(x, y, z) = (-30.398, -22.48, 61.338)$ and was run until $t = 50$.

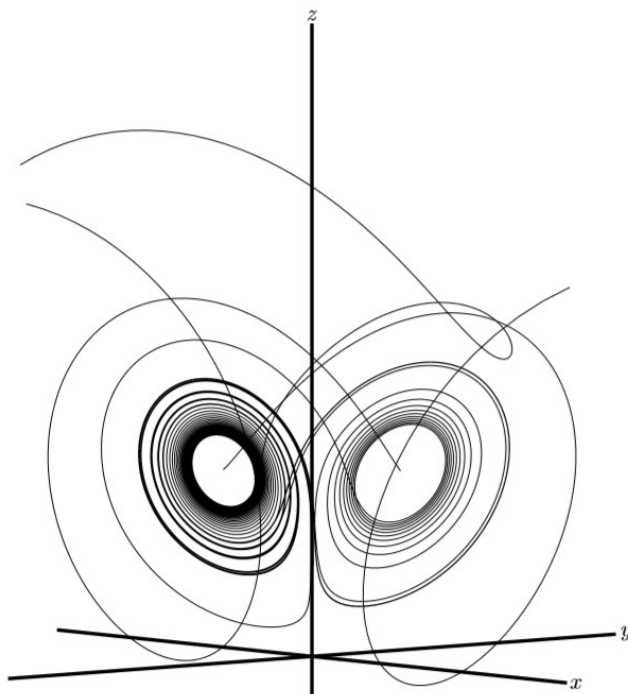


9.3.4 (cont.)

This simulations have initial conditions $(x, y, z) = (62.011, -9.6311, 51.332)$ and $(-63.425, 6.1968, 52.604)$ and were run until $t = 5$. The linearization predicts a stable eigendirection and a stable spiral, so there is a tiny oscillation at the end. (We only show the last plot since the others are uninteresting.)



And now all three together.



9.3.8

a)

Yes. $\dot{r} = r(1 - r^2)$ has fixed points at $r = 0, 1$ so if r starts in $[0, 1]$ then r is trapped there forever. Therefore D is an invariant set.

b)

Yes. The fixed point $r = 1$ of $\dot{r} = r(1 - r^2)$ is stable, so every open set is attracted to D , which makes $\mathbb{R}^2$ the basin of attraction.

c)

No. Any proper subset of D containing $x^2 + y^2 = 1$ is invariant and attracts an open set of initial conditions.

d)

Yes. The set $x^2 + y^2 = 1$ is an attractor from part (c), and $\dot{\theta} = 1$ guarantees $x^2 + y^2 = 1$ is minimal since all the points in $x^2 + y^2 = 1$ are visited within 2π time for an initial condition that starts on the circle.

9.3.9

The following graph was made by running an arbitrary initial condition until the trajectory was on the Lorenz attractor at $(x, y, z) = (0.8961, 1.5605, 11.246)$. Then the trajectory and a perturbed trajectory of $(x, y, z) = (0.8961, 1.5605, 11.246 + 0.001)$ were simulated until $t = 25$ and the error computed in the following graph. Then we used a linear fit of the error between $t \in [1, 10]$ so as to avoid the initial transient and leveling off of the error. The slope is an estimate of largest Liapunov exponent and was $\lambda = 0.8623 \approx 0.906$, the latter being the “real” value calculated with more advanced methods.

